IS1203: Foundations of Information Systems

Topic 3:
Different Types of Business IS

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Intended Learning Outcomes

After completion of this topic, you'll be able to,

- Identify different types of business information systems in an organization.
- Categorise IS from functional perspective and constituency perspective.
- Analyse functionalities of different types of business information systems.
- Draw dataflow diagrams to depict business processes and describe functionalities of a BIS.
- Describe different types management reports.

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Types of Business Information Systems

1. ***A functional perspective:*** Identifying *systems* by their major **business function**.
   i. Sales and Marketing System
   ii. Manufacturing and Production System
   iii. Finance and Accounting System
   iv. Human Resource System
2. ***A constituency perspective:*** Identifying *systems* in terms of the major **organizational groups** that they serve.
   i. TPS
   ii. MIS
   iii. DSS
   iv. ESS

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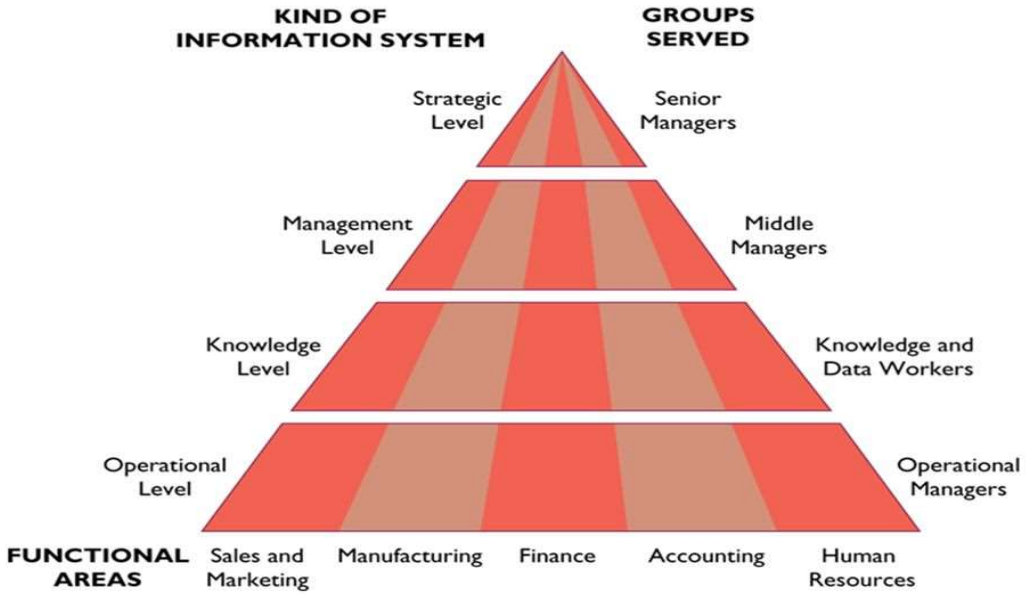
IS Requirements



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Sales and Marketing Information System

- **Marketing business processes**
  - Identifying customers,
  - Introducing products and services to meet their needs
  - Promoting products and services
- **Sales processes**
  - Selling the products and services
  - Taking orders
  - Contacting customers
  - Providing customer support (e.g., after-sales services)
  - Introducing promotions- new offerings and discounts
  - Generating reports on time, region,…etc.

Sales and Marketing Information System

- **Strategic Level**
  - Monitors trends affecting new products, sales opportunities and competitors' performance.
  - Supports new products and services
- **Management Level**
  - Supports market research, advertising and promotional campaigns and pricing decisions
  - Analyzes sales performance of the sales staff.
- **Knowledge Level**
  - Supports analyzing marketing data.
- **Operational Level**
  - Assists in locating and contracting prospective customers, tracking sales, processing orders, and providing customer service support.

Sales and Marketing …contd.

Major functions:

- Sales management, market research, promotion, pricing, identifying new products

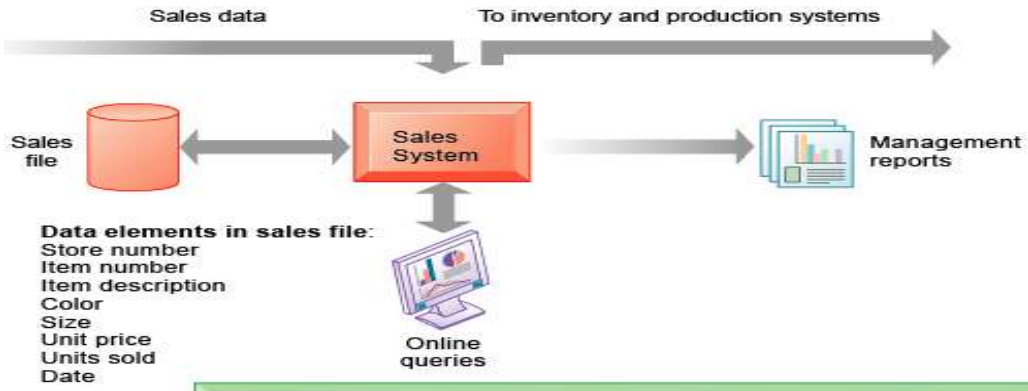
Major application systems:

- Sales Order Information System,
- Market Research System,
- Pricing System

Sales and Marketing …Contd.

System	Description	Organizational level
Order Processing	Enter, process, and track orders	Operational
Price Analysis	Determine prices for products and services	Management
Sales Trend Forecasting	Prepare 5-year sales forecasts	Strategic

E.g. A Sales System



Sales Report: Week Ending 2/7/07

Item No.	Description	Color	Size	Unit Price	Units Sold	Total Sales
294	Sports bag	Black	Small	10.00	10,451	$104,510
295	Sports bag	Black	Medium	20.00	21,800	$436,000
394	Sports bag	Red	Small	10.00	5,331	$53,310

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Manufacturing and Production Information Systems

- **Strategic Level**
  - Deal with the firm's long-term manufacturing goals, such as where to locate new plants or whether to invest in new manufacturing technology
- **Management level**
  - Analyze and monitor manufacturing and production costs and resources
- **Knowledge level**
  - Create and distribute design knowledge or expertise to drive the production process
- **Operation level**
  - Deal with the status of production tasks

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Manufacturing and Production Systems

Major functions of systems:

- Scheduling, purchasing, shipping, receiving, engineering, and operations

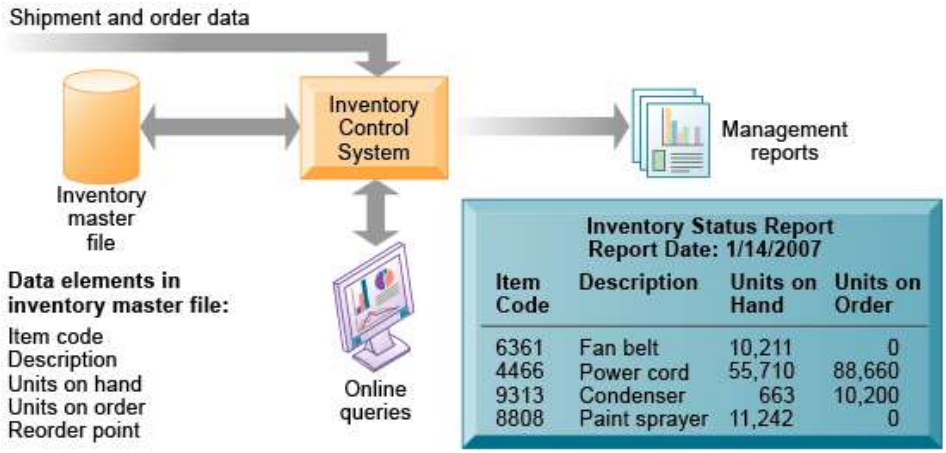
Major application systems:

- Materials Resource Planning Systems,
- Purchase Order Control Systems,
- Engineering Systems,
- Quality Control Systems

Manufacturing and Production Systems

System	Description	Organizational Level
Machine Control	Control the actions of machines and equipment	Operational
Production Planning	Decide when and how many products should be produced	Management
Facility Location	Decide where to locate new production facilities	Strategic

E.g., Inventory Control System



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Finance and Accounting Systems

Strategic Level

- Establish long-term investment goals for the firm
- Provide long-range forecasts of the firm's financial performance

Management level

- Help managers oversee and control the firm's financial resources

Knowledge level

- Support finance and accounting by providing analytical tools and workstations for designing the right mix of investments

Operation level

- Track the flow of funds in the firm through transactions such as paychecks, payments to vendors, and receipts.

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Financing and Accounting Systems

Major functions:

- Budgeting, general ledger, billing, cost accounting

Major application systems:

- Updating /Handling General Ledger,
- Accounts Receivable,
- Accounts Payable,
- Budgeting,
- Funds Management Systems

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Financing & Accounting Systems…Contd.

System	Description	Organizational Level
Accounts Receivable	Tracks money owed to the firm	Operational
Budgeting	Prepares short-term budgets	Management
Profit Planning	Plans long-term profits	Strategic

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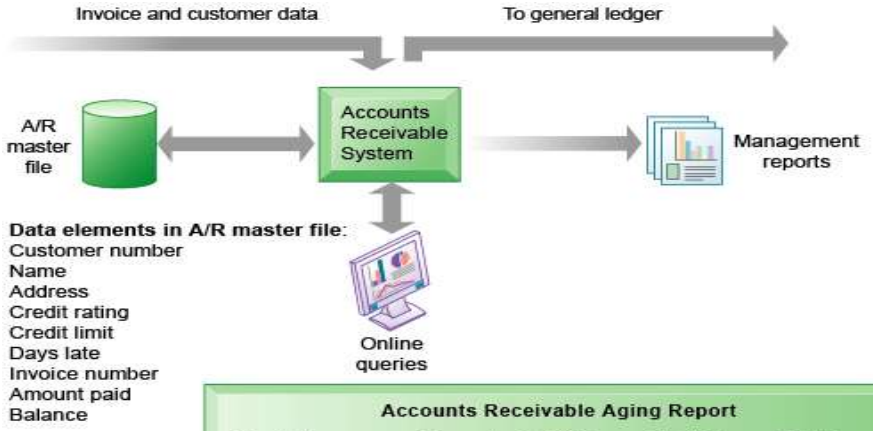
E.g. An Accounts Receivable System



Data elements in A/R master file:
Customer number
Name
Address
Credit rating
Credit limit
Days late
Invoice number
Amount paid
Balance

Accounts Receivable Aging Report

Cust. No.	Name	Current Balance	1–30 Days Past Due	31–60 Days Past Due	61+ Days Past Due	Total
5043	Myers Co.	0	500.00			500.00
6219	JT Garden	1,500.00		700.00		2,200.00
6932	Best Home	1,000.00				1,000.00

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Human Resource Systems

- **Strategic Level**
  - Identify the manpower requirements (Skills, education level, types of positions, number of positions, and cost) for meeting the firm's long-term business plans
- **Management Level**
  - Help managers monitor and analyze the recruitment, allocation, and compensation of employees.
- **Knowledge level**
  - Support analysis activities related to job design, training, and the modelling of employee career paths and reporting relationships.
- **Operation level**
  - Track the recruitment and placement of firm employees.

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Human Resource Systems

Major functions of systems:

- Personnel records, benefits, compensation, labor relations, training

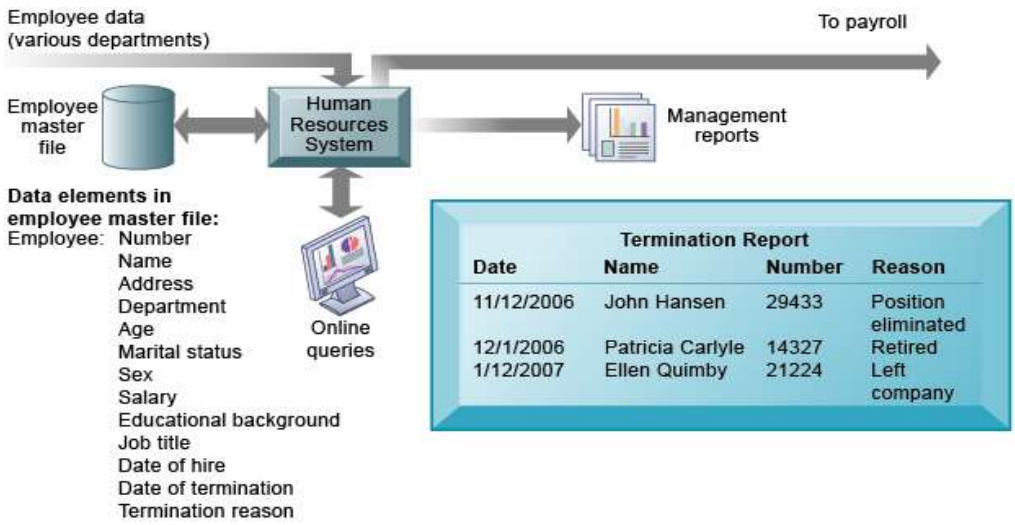
Major application systems:

- Payroll
- Employee records handling systems
- Benefit systems
- Career path systems
- Personnel training systems

Human Resource Systems (Cond.)

System	Description	Organizational Level
Training and development	Tracks employee training, skills, and performance appraisals	Operational
Compensation analysis	Monitors the range and distribution of employee wages, salaries, and benefits	Management
Human resources planning	Plans the long-term labour force needs of the organization	Strategic

Human Resource Systems…contd.



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Activity 1

Describe the information systems supporting the following major business functions.

i. Sales and marketing
ii. Manufacturing and production
iii. Finance and accounting
iv. Human resource management

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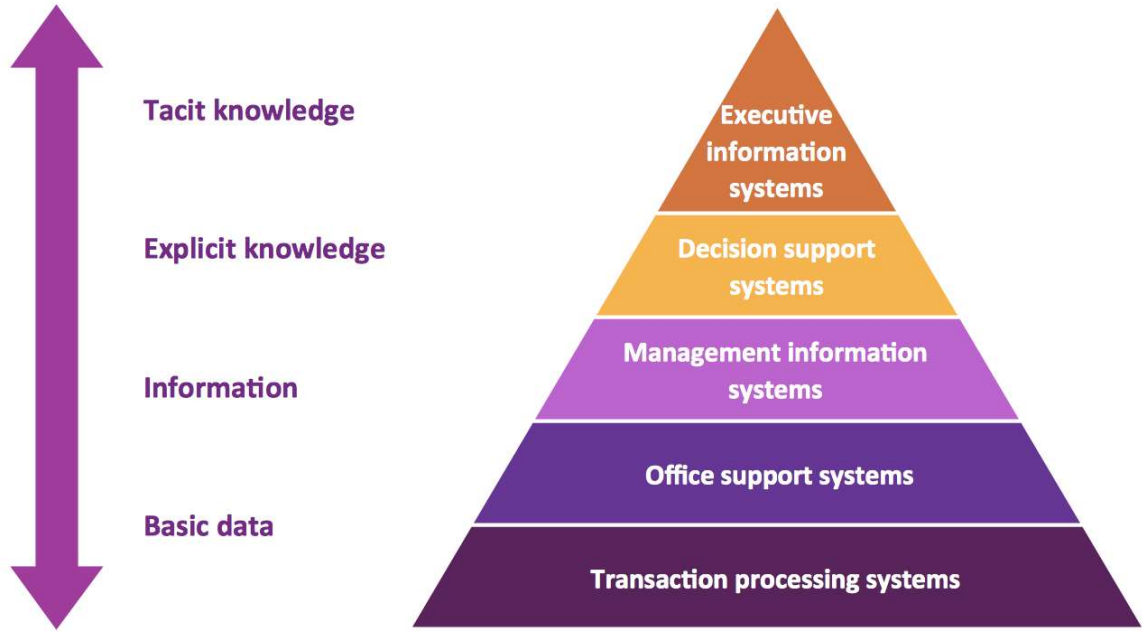
Activity 2

- Prepare a table analysing the functionalities of different types of functional information systems in a business organization.

Major Types of Business Information Systems: Constituency Perspective

1. Transaction Processing Systems (TPS)
2. Office Automation Systems (OAS)
3. Management Information Systems (MIS)
4. Decision Support Systems (DSS)
5. Executive Support Systems (ESS)



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Transaction Processing System (TPS)

- Basic information system that process and produce reports
- Serves the Operational Level of an organization
- automates repetitive information processing activities to;
  - increase speed, volume of data, accuracy and to
  - lower the cost of processing each transaction, so it makes the organization more effective.
- a highly structured system

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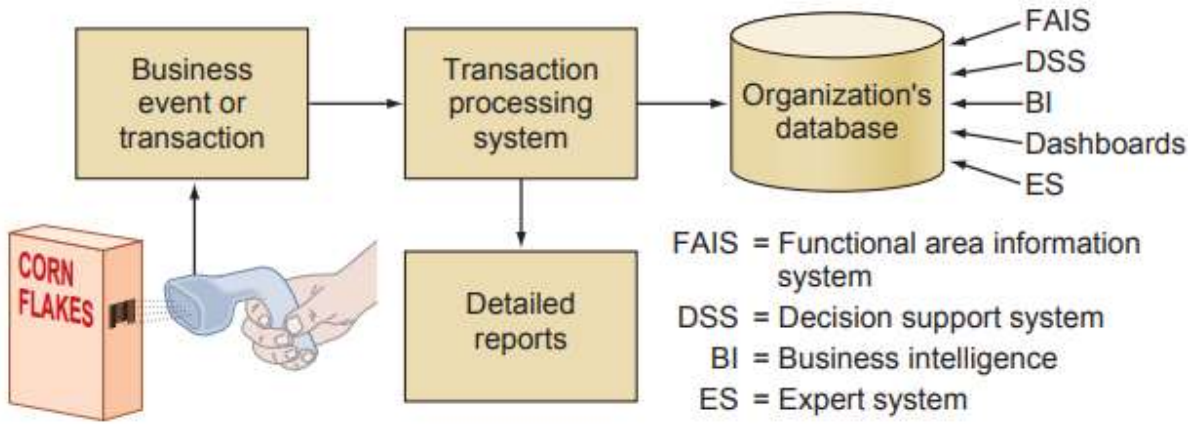

How transaction processing systems manage data.

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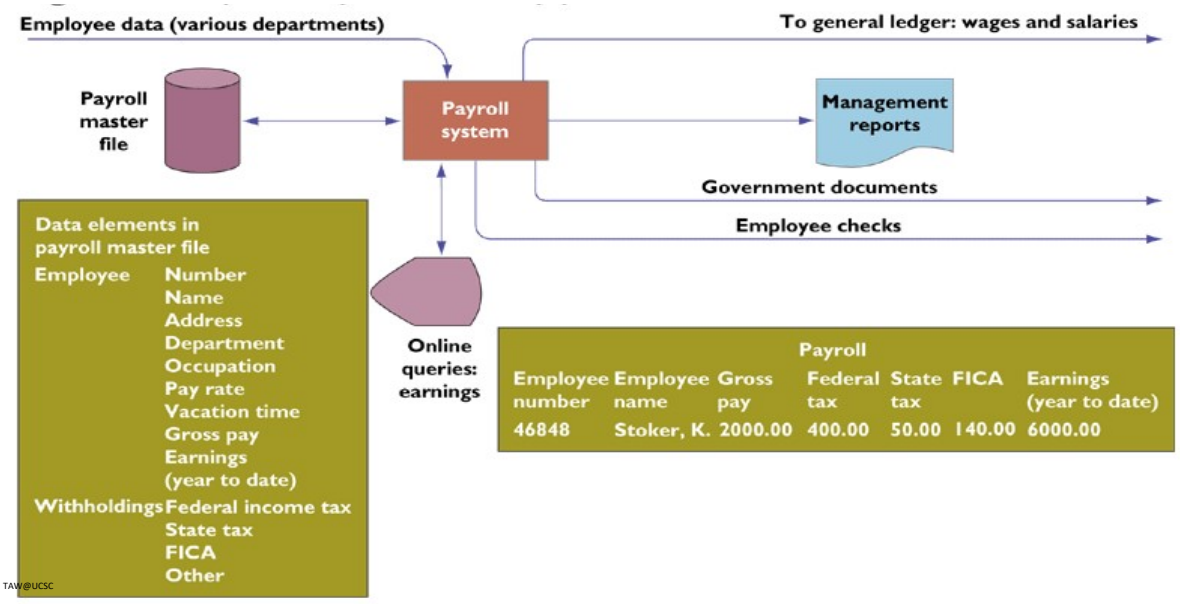
E.g., A Symbolic Representation for a Payroll



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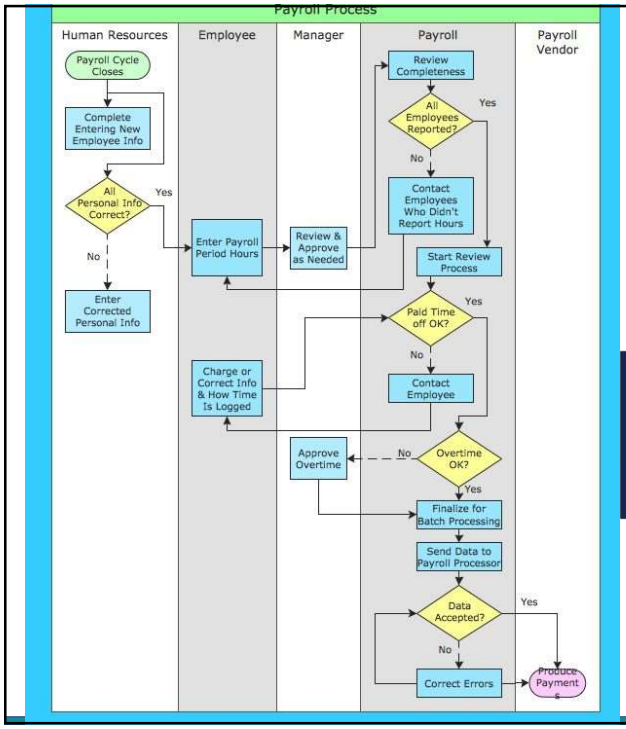

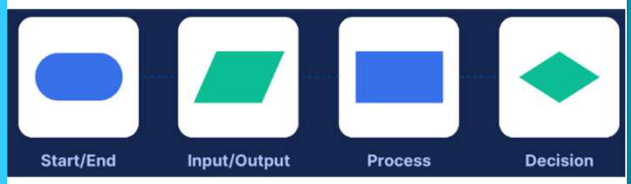

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Activity

- Study the functionalities of the Payroll TPS and identify the inputs, outputs, processes and type of data to be stored.

Inputs	Processes	Outputs

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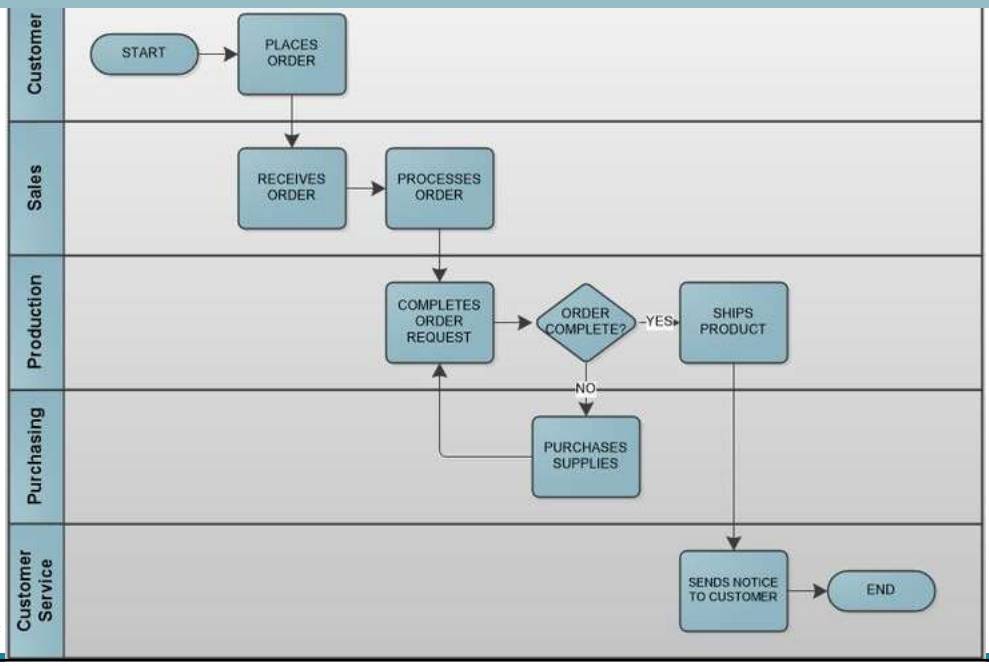
Order Processing Process



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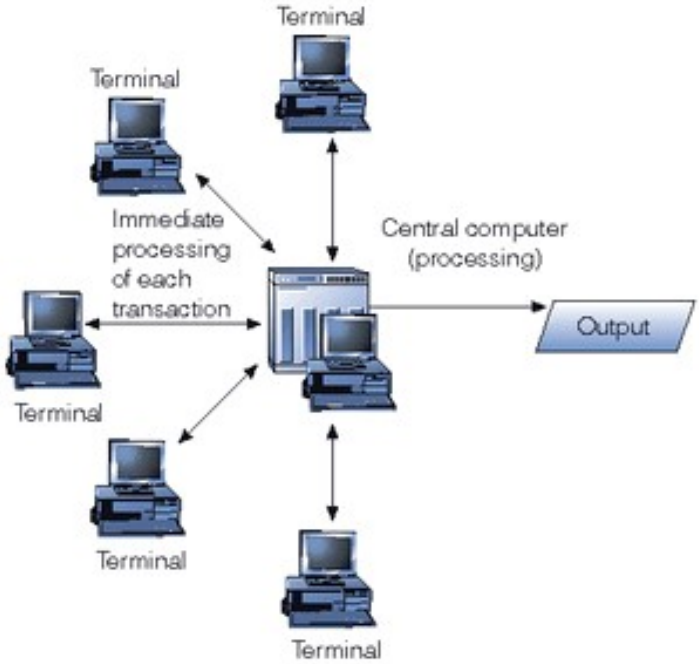
Real-Time Processing



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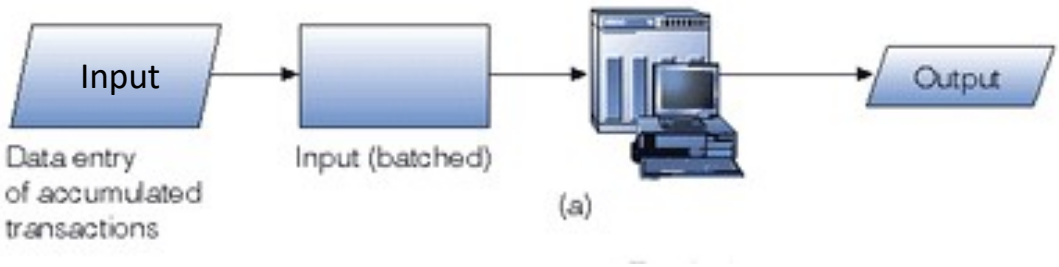
Batch Processing



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Batch vs. Real Time Processing

Batch Processing	Real-Time Processing
The collection and storage of data, for processing at a scheduled time when a sufficient amount of data has been accumulated	The immediate processing of data after the transaction occurs, with the database being updated at the time of the event
Transactions Collected and Organised into Batches → Transaction File created → Transaction File stored → Master File updated at scheduled time periods	Transaction event occurring ⇄ Online computer database updating
Examples: • Cheque Clearing • Generation of Bills • Credit Card Payments	**Examples:** • Reservation Systems • Point of Sales Terminals (POS)
Advantages / Disadvantages: • Many transactions are completed at one time in a single process • Data takes time to be processed	**Advantages / Disadvantages :** • Data is processed immediately • The act of processing data is repetitive

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Features of a TPS

- Exist in the operational level
- Automation of routine transactions
- Central for business tasks
- Acts as data source for other information systems such as MIS
- Manages large amount of data
- Improves Efficiency and
- Increase Competitive Advantage

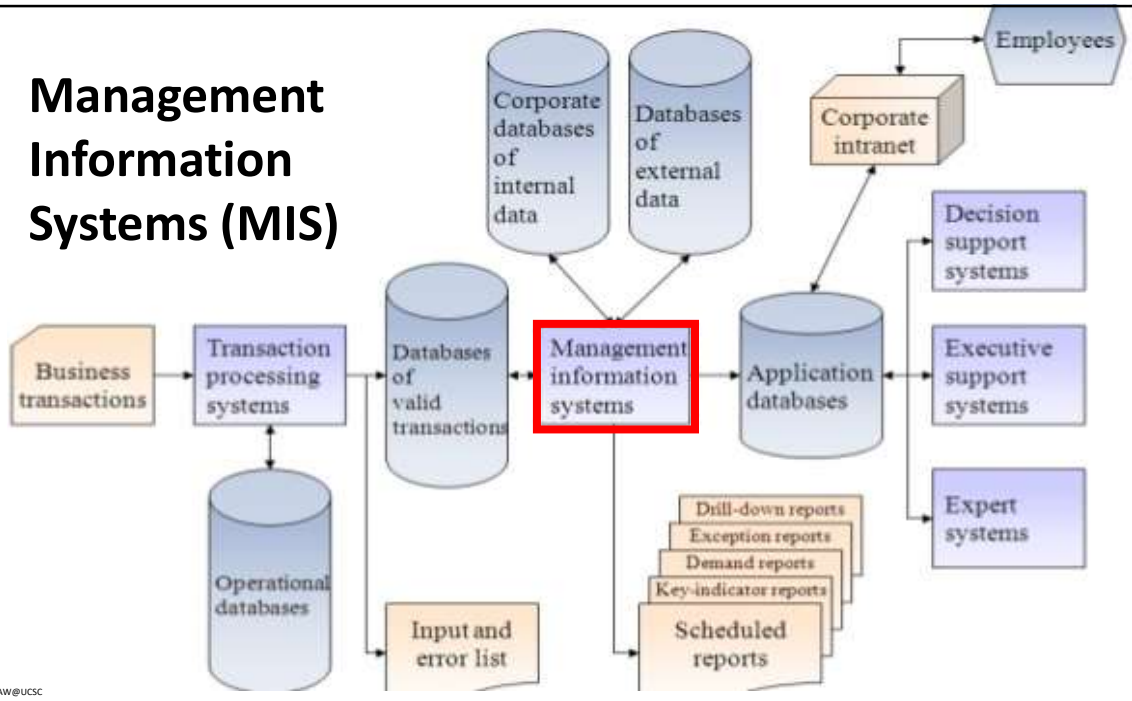
Office Automation Systems (OAS)

- An office system that **collects, processes, stores, manipulates and transmits electronic data and office information** needed for accomplishing office tasks.
- The **backbone of office automation is a LAN,** which allows users to transfer data, send mails, conduct presentations, and discussions across a network.
- Targeted at **meeting the knowledge needs of *data workers*** within an organisation
- Typical OAS **handle and manage documents, schedules and communication.**

Examples:

- Office applications
  - *MS office*
  - *Open Office*
- Accounting software- *QuickBooks*
- Help managing customers and employees- *Receptionist software*
- Collaborative work – *MS Teams, Google Drive, Workspaces, Flock*
- Manage social media
  - **Hootsuite** - All-in-One Social Media Tool supporting social network integrations for Twitter, Facebook, Instagram, LinkedIn and YouTube
  - **Buffer** - All-you-need social media toolkit for small businesses and marketing teams to schedule posts, analyze performance, and manage their accounts in one place
  - **MeetEdgar** - Simplifies social media automation by using smart evergreen content, categorization, time slots, and auto-import features.

Management Information Systems (MIS)



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Management Information Systems (MIS)

- MIS is an IS to monitor and control business and predict the future performance
- Generate reports on the company's basic operations using data supplied by TPS.
- Provide middle managers with reports on organization's current performance.
- Provide answers to routine questions
- Focus entirely on internal events, providing the information for short-term planning and decision making.

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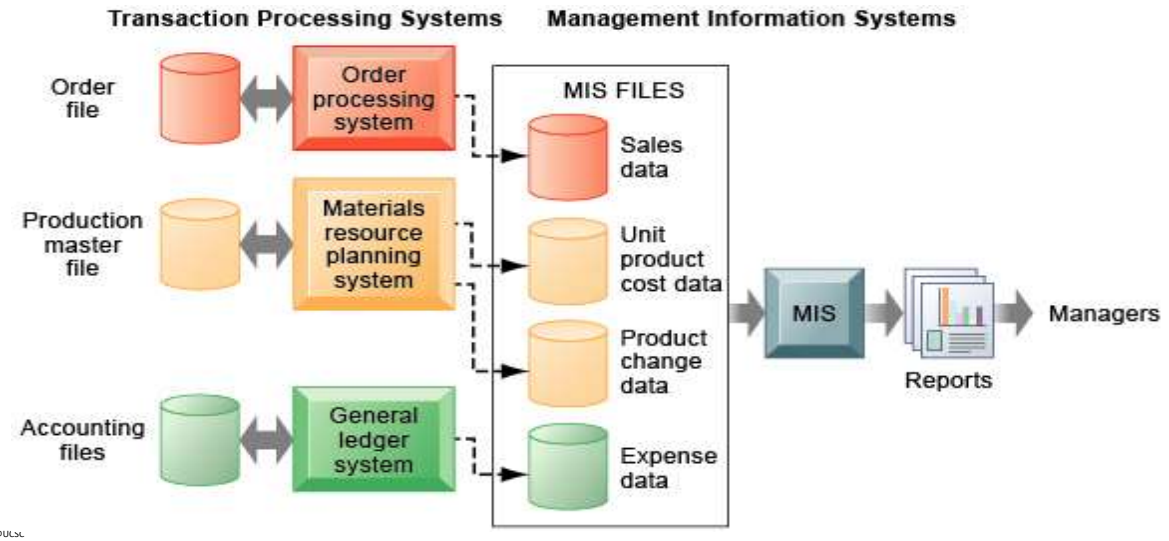
About MIS

- Structured and semi-structured decisions
- Report control oriented
- Designed to report on existing operations
- Internal orientation
- Have little analytical capability

TPS & MIS Relationship



Management reports

- **Scheduled reports-** produced periodically, such as daily, weekly or monthly
- **Key – indicator reports** -summarizes the previous day's critical activities to support taking quick action
- **Demand reports** -provide certain information upon request
- **Exception reports** - used to record details of any adverse situation arising
- **Drill down reports** - provide specialized reporting on hierarchies you have set up

Sample MIS Report: Sales

ACCOUNTING SALES REPORT

Store Location : **Store** Date : **May 1, 2**

Total Sales: **$9,263.58** (Less Voids, Promos, Discounts, Taxes, Add Charges)

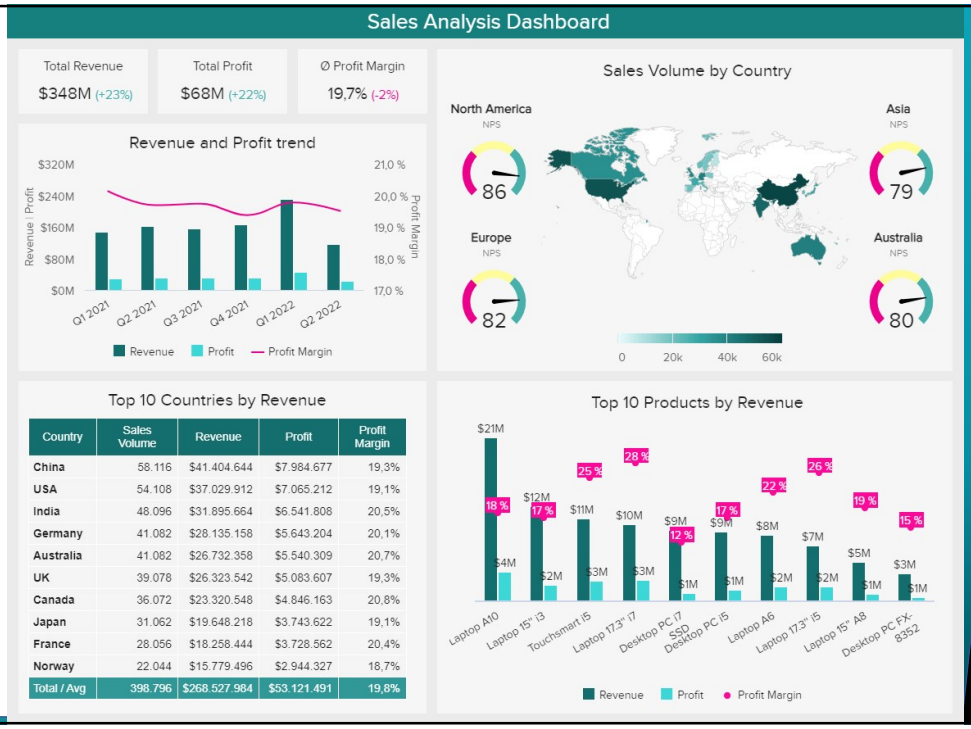
SALES BY DEPARTMENT

Sales Category	Net Sales	Gross Sales	Refunded	Discounts	Taxes	Cost
None Department	$7,896.50	$8,248.88	$59.50	$160.00	$132.88	$223.00
Books	$1,249.08	$1,369.44	$36.13	$3.40	$80.84	$483.00
Outdoor Supplies	$76.00	$80.76	$0.00	$0.00	$4.76	$22.00
Accessories	$42.00	$44.63	$0.00	$0.00	$2.63	$34.00
Totals:	**$9,263.58**	**$9,743.71**	**$95.63**	**$163.40**	**$221.11**	**$762.00**

SALES BY CUSTOMER GROUP

Group Name	Net Sales	Gross Sales	Capture
NOT LOGGED IN	$6,129.00	$6,358.66	$6,358.6
General	$1,571.38	$1,720.30	$1,720.3
VIP Customers	$1,400.00	$1,487.50	$1,487.5
Wholesale	$102.00	$108.39	$108.39
Retailer - 10%	$61.20	$68.86	$68.86
Totals:	**$9,263.58**	**$9,743.71**	**$9,743.7**

Digital Dashboard



Country	Sales Volume	Revenue	Profit	Profit Margin
China	58.116	$41.404.644	$7.984.677	19,3%
USA	54.108	$37.029.912	$7.065.212	19,1%
India	48.096	$31.895.664	$6.541.808	20,5%
Germany	41.082	$28.135.158	$5.643.204	20,1%
Australia	41.082	$26.732.358	$5.540.309	20,7%
UK	39.078	$26.323.542	$5.083.607	19,3%
Canada	36.072	$23.320.548	$4.846.163	20,8%
Japan	31.062	$19.648.218	$3.743.622	19,1%
France	28.056	$18.258.444	$3.728.562	20,4%
Norway	22.044	$15.779.496	$2.944.327	18,7%
Total / Avg	398.796	$268.527.984	$53.121.491	19,8%

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Types of function-specific MIS

- Marketing Information System
- Human Resource Information System
- Financial Management Information System
- Production Management Information System

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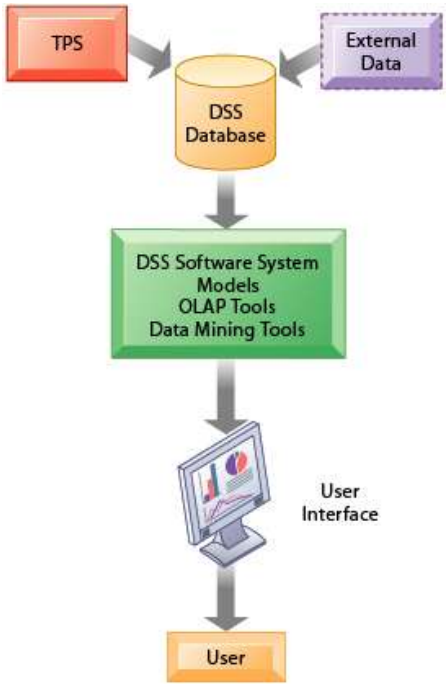
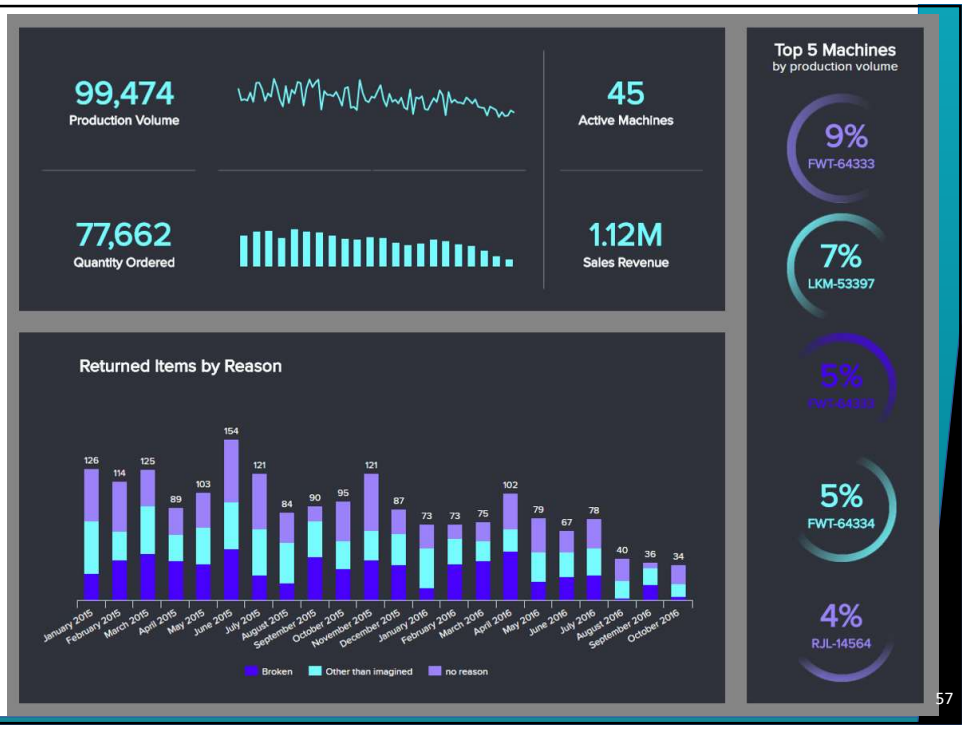
Decision Support Systems (DSS)

- Are used by managers (super users) and business analysts
- Focuses on problems that are unique, rapidly changing, and of which the procedure of addressing is not specified in advance
- DSS answered questions such as
  - **What would be the impact on production schedules if we were to double sales in the month of December?**
  - **What would happen to our return on investment if a factory schedule were delayed for six months?**
- Use both internal **information from TPS and MIS** and external information such as **current stock prices or product prices of competitors**
- Have **more analytical power** than other systems.

Components of DSS



Production Monitoring



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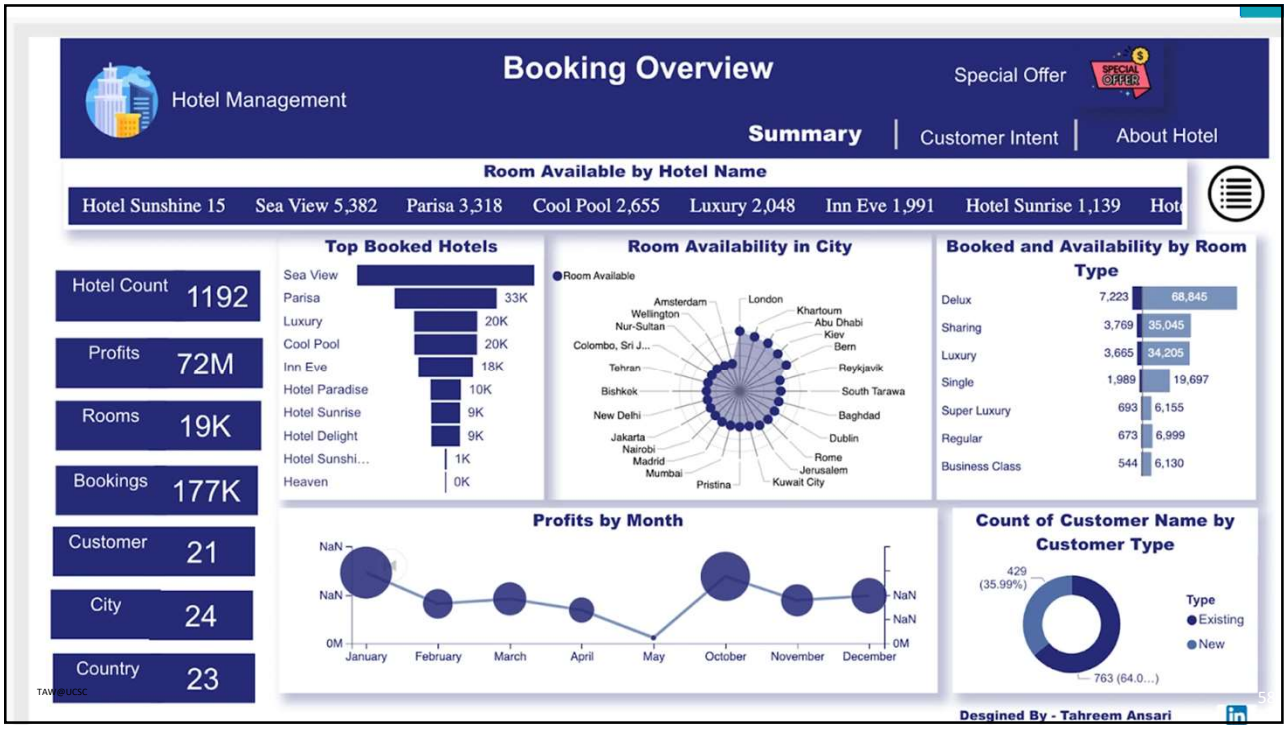

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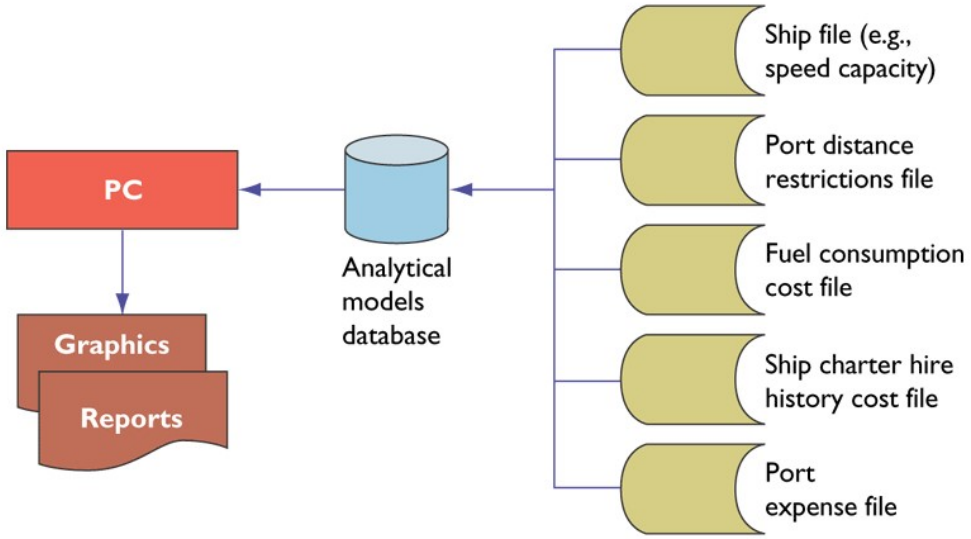
e.g., A Voyage-estimating DSS



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Review Exercise

- What is the difference between Exception Reports and Key-indicator reports?

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Executive Support Systems (ESS)

- Are used by Senior Management who make decisions at the strategic level of the organization
- Often information is delivered through a portal
- Addresses **non routine decisions** requiring judgment, evaluation, and insight
- There is **no agreed-on procedure for arriving at decisions**
- Being designed to incorporate data about external events such as new tax laws or competitors and draw summarized information from internal MIS and DSS

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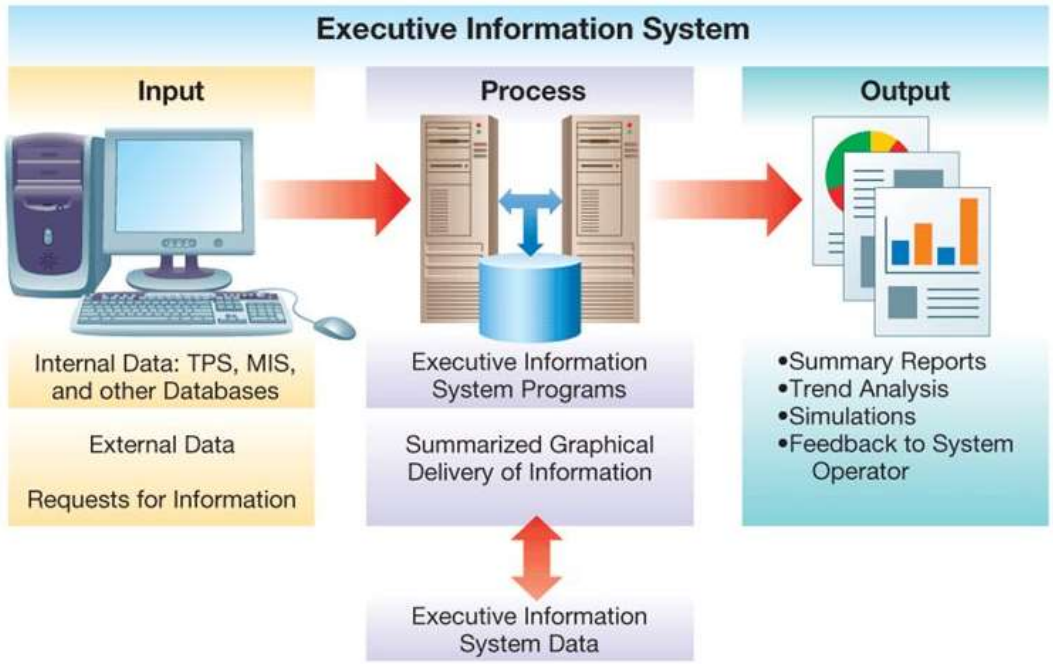

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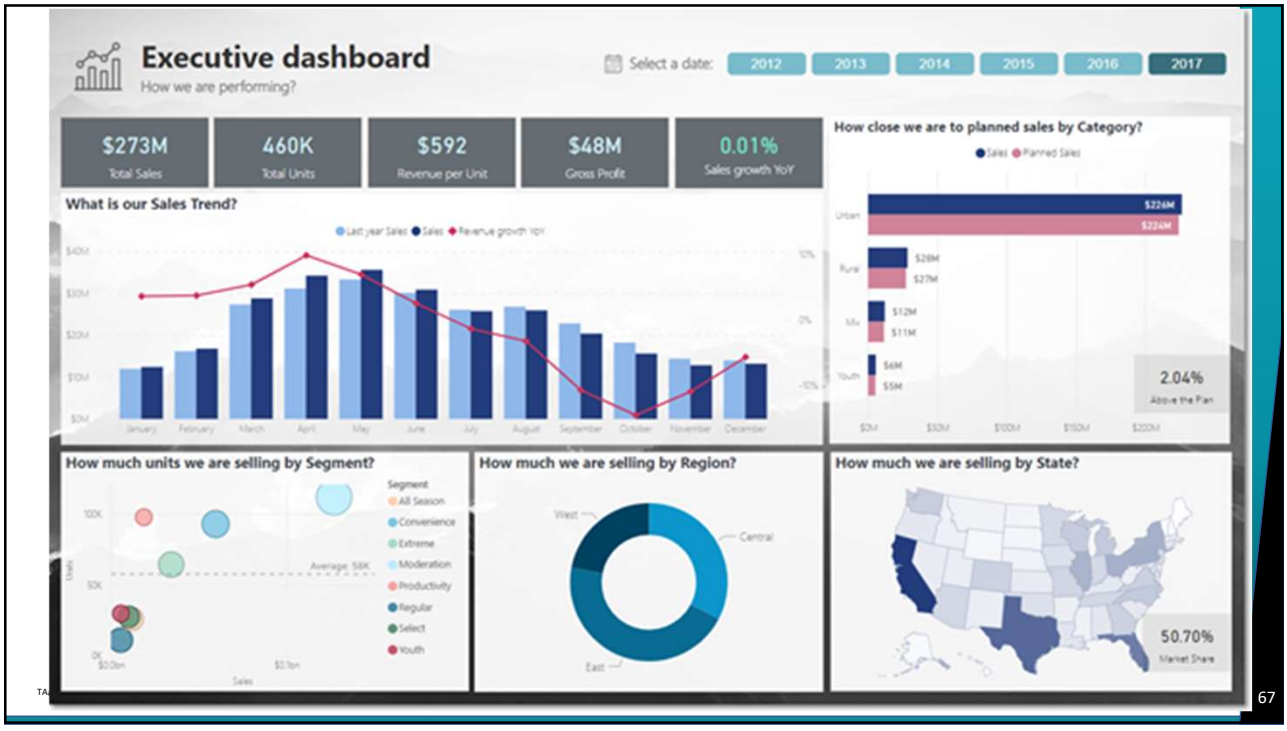
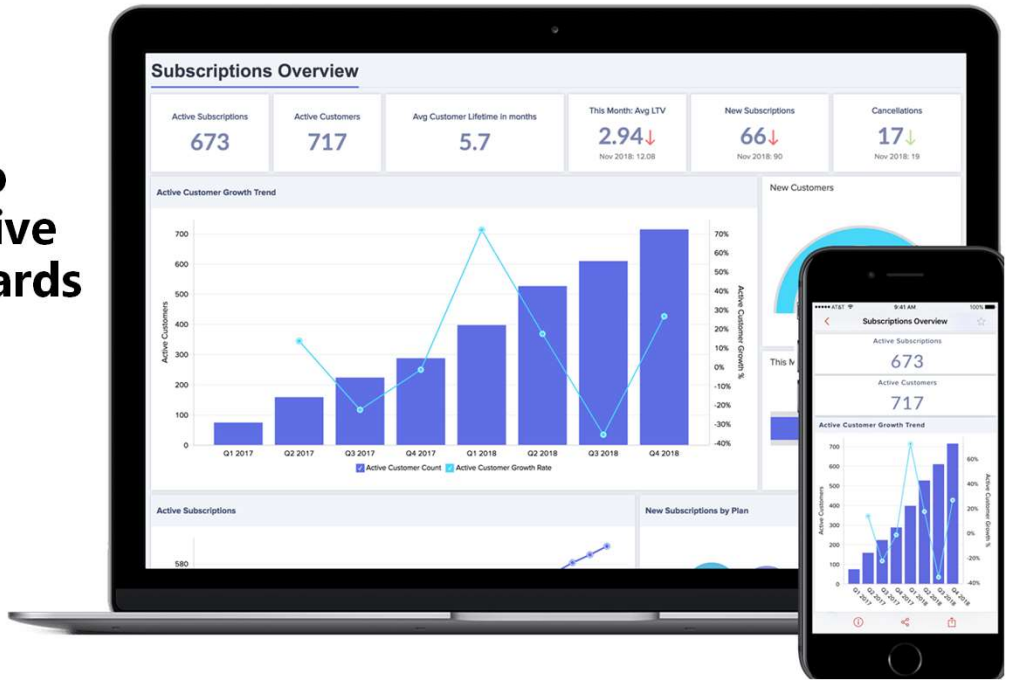
About ESS

- Use by Top level management
- Designed for individuals
- Ties CEO to all levels
- Sophisticated data analysis and modelling tools
- Flexibility
- Quick response time



Zoho Executive Dashboards



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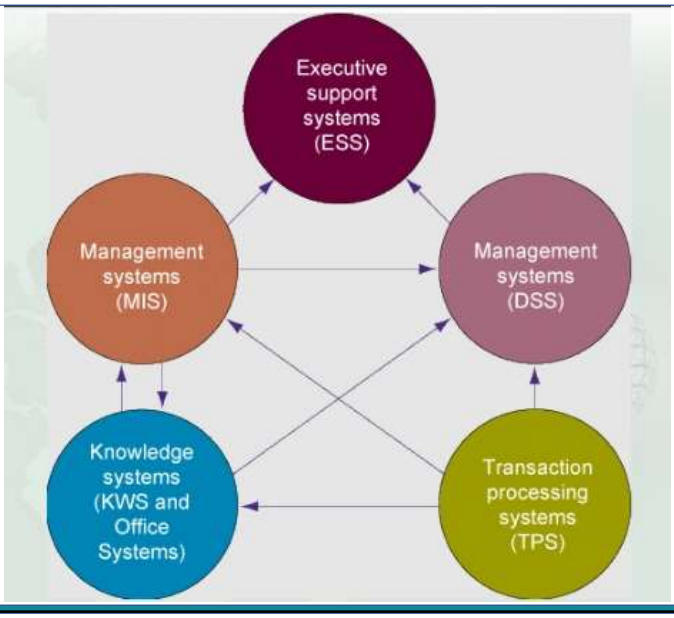
Interrelationship among Systems



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Exercises

- Compare DSS and ESS
- Prepare a table analysing the functionalities of different types of constituency perspective systems in a business organization.

Summary

1. Sales and Marketing System
2. Manufacturing and Production System
3. Finance and Accounting System
4. Human Resource System
5. Transaction Processing Systems (TPS)
6. Office Automation Systems (OAS)
7. Management Information Systems (MIS)
8. Decision Support Systems (DSS)
9. Executive Support Systems (ESS)